

FURTHER WRESTLING WITH RECESSED-CAN LIGHTS

How do you air seal IC-rated light fixtures that aren't accessible from above? Do the air sealing from below.

by LARRY ARMANDA

Since my first article on recessed-can lighting (“A Recessed Can of Worms,” *HE* Jan/Feb '01, p. 42), there have been great strides in the development of new and improved insulated ceiling airtight (ICAT) recessed-can light fixtures for new and remodel construction. And new compact fluorescent models have become more available for reducing electricity use. The driving force for these developments may be new building codes mandating the use of the safer and more efficient lighting. What concerns me as a private consultant is how to deal with standard IC-rated recessed-can lights that were installed in cathedral ceilings or attic spaces that are not accessible and therefore cannot be air sealed from above. Standard IC-rated fixtures are still being installed in new construction, as well as in remodel projects, by some contractors and by homeowners themselves. (Non IC-rated lighting fixtures are another story all together.)

Yes, I still get calls from homeowners every winter when they notice holes in the snow on their roofs above the locations of all their recessed-can light fixtures. Even better yet, the calls I truly love: “Why is my house the only house in the neighborhood that never has any snow on the roof? Do you think it has anything to do with the thirty-six recessed-can lights I recently installed?” Well, it just may have something to do with it, or maybe the aliens have grown tired of crop circles! During the winter of 2004/2005, that is exactly what happened with me and my business (not the aliens-on-the-roof thing). To try to address the problem

of inaccessible can lights, I’ve gone back to the lab and into the field to test some ideas and different commercially available retrofit kits (see “First a Little History”).

In the Lab

How do you make a standard IC-rated recessed-can light fixture more airtight when you can’t build an air-sealed box from above? What air sealing material can you use to withstand the temperatures expected inside the fixture? What materials will withstand summer temperatures of 140°F? What lamp is the best choice to keep the thermal overload from shutting the fixture off? How much air reduction can we expect after air sealing these fixtures? How cost-effective are manufactured retrofit kits?

To answer these questions I started back in the lab with my original test

mockup (see photo above). I removed the 13 inch x 16 inch x 10 1/4 inch air-sealed drywall box from over the top of the fixture, attached a thermal sensor to the side of the fixture, and placed R-19 fiberglass batt insulation over and around the fixture. I placed a second sensor above the insulation to record the ambient temperature and then placed the original 28 inch x 53 inch x 24 1/2 inch box made with insulation board over the entire mockup. The inside of the insulated box was heated with a 300 watt lamp controlled by a Chromolox thermostat, in order to simulate a hot ambient attic air temperature of 140°F. All the tests were performed at this summer ambient air temperature. I disabled the thermal cutout switch on the fixtures before testing them.

The lamps I chose for the test are a 100W A-base lamp, a 90W Capsylite



This mockup was used to test recessed-can lights. The 300W lamp on the left was used to simulate an attic temperature of 140°F.

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flood, a 20W compact fluorescent flood, and a 20W triple biac compact fluorescent. I chose the 100W A-base lamp to start the testing, since I expected this to be the worst lamp of the group. The temperature of the fixture, after three hours of operation, went to 310°F—well over the 194° limit of the thermal

above the thermal cutout switch rating. The good news is that the new wire nut didn't burn off this time and the metal foil tape was still intact. I did think of calling my insurance agent, though, to check my coverage. The 20W compact fluorescent flood lamp raised the fixture temperature to 144°F after five hours of

our light fixtures, I conducted a blower door/pressure pan test on the four IC-rated recessed can lights above the kitchen countertop. These fixtures are in an inaccessible attic space that is insulated with cellulose insulation. The trims are the open black baffle type. The fifth recessed can I tested at home is an uninsulated eyeball that is located in the center of the kitchen, between the first and second floors. I tested three additional recessed-can fixtures in the first-floor master bathroom. The gasketed glass-trim shower light and two open trims in the bathroom are insulated with cellulose insulation.

I depressurized the house and did pressure pan tests over each fixture. After documenting all the pressure pan readings, I removed all the open black baffle trims from the recessed-can fixtures in the kitchen and sealed all of the holes with the same aluminum duct tape that I had used in the lab. In all but one of the fixtures, I installed the tape on the inside of the fixture and on the drywall just under the reveal of the trim so that it would not be seen. I used tape to seal the holes inside one of the fixtures above the kitchen countertop, but I didn't seal it to the drywall, as I did with all the other fixtures. All the fixtures were equipped with 75W reflector flood lamps. (See Table 2 for the air flow around the fixtures before and after air sealing.)

The results of the pressure pan tests indicate that a substantial reduction of leakage is possible by simply taping all the holes inside the fixture as well as tap-

Table 1. Temperature Data of Air Sealed Light Fixture

Lamp Type	Lamp Wattage	Maximum Fixture Temperature (°F)	Operation Time (Hours)	Comments
	100W A-base	310	3	Wire nut burned off thermal jumper
	90W Capsylite	217	3	
	20W fluorescent flood	144	5	
	20W fluorescent biac	162	6	

cutout switch that I had disengaged. The wire nut that I used to jumper out the thermal cutout switch was on the floor; it had melted off the connection (see photo on p. 33). The good news was that the metal foil tape held in place.

After returning from the store with a new fire extinguisher, I started to test the next lamp, the 90W Capsylite flood. After three hours of operation the fixture reached a temperature of 217°F, also

operation. This is a good thing. The 20W triple biac compact fluorescent produced a fixture temperature of 162°F after six hours of operation (see Table 1).

Duct Tape and Air Leakage At Home

Using my home as the next test site, and after trying to convince my wife that I was only cleaning the insides of

First a Little History

My original research project with can lights was to record the temperatures of the lights when the light fixtures were installed under an air-sealed box made of 1/2-inch drywall as per the guidelines of the International Energy Conservation Code 2000 and the National Electrical Code. I tested seven different lamps and wattages at different ambient (attic)

temperatures. No thermal insulation was installed over the air-sealed box. Temperature sensors were installed at different locations of the drywall box and on the light fixture itself. The thermal overload switch was disconnected and a wire nut used to jumper out the thermal switch to keep the fixture from shutting off if temperatures exceeded the 194°F

limit of the thermal switch. The highest temperature recorded was the result of using the 100W standard A-base lamp—the temperature reached 185°F. This was recorded on the drywall box directly above the fixture, with an air temperature of 180°F inside the drywall enclosure. Inside the electrical terminal box, the recorded temperature was 164°F.

ing the fixture to the drywall. Also, there was no reduction of leakage through the fixture above the kitchen countertop that I sealed inside the fixture but not to the drywall.

This air-sealing method was very time consuming. The biggest problem with this method is trying to reattach the small springs that hold the trim to the ceiling. I found that I had to stretch the spring to attach it to the lamp bracket that holds

Table 2. Pressure Pan Readings of Recessed-Can Lights with House Depressurized to -50 Pa Before and After Sealing

Fixture Location	Pressure Pan Before Sealing (Pa)	Pressure Pan After Sealing (Pa)	Comments
Kitchen by stove #1	-41	-41	Not sealed to ceiling drywall
Kitchen corner #2	-40	-17	Sealed to drywall
Kitchen above sink #3	-38	-10	Sealed to drywall
Kitchen right of sink #4	-38	-19	Sealed to drywall
Kitchen eyeball	-33	-15	Sealed to drywall
Bath shower light	- 4	- 4	No sealing was done
Bath light	-22	- 8	
Bath light	-21	-10	

Table 3 Results of Pressure Pan Test for Retrofit Kits

Trim	Manufacturer	Pressure Pan Reading (Pa)	Cost (\$)	Comments
	6-in plastic opal trim and lens by Lithonia	-48	13	Trim did not have a gasket around trim flange and leaked air between trim and drywall. No reduction of air flow.
	Unknown	-18	Unknown	Trim had a heavy closed-cell-rubber-type gasket which appeared to seal very well at the drywall.
	Air Tite Super Trim by Halo	-30	14	Trim did not have a gasket around trim flange but did have a gasket at the lamp socket. Trim leaked air at drywall. This trim was marked Washington State Approved. Slight reduction of air flow.
	Builders Best	-24	34	Retrofit kit that included a highly reflective interior, a compact fluorescent lamp, and a vented lens cover. The retrofit kit was dimmable and had a foam gasket around trim. Universal mount to Halo and Juno fixtures. Trim leaked air at drywall.
	Air-Tite Kit by Lithonia Lighting	-25	2.35	Kit consisted of foam rubber gasket material to be installed inside the fixture as well as around the trim. It was easy to install and was enough to seal all of the holes in the fixture. The air leakage appeared to be inside the fixture as well as around the trim.

the lamp socket. But the work paid off. The tested kitchen lights, which are equipped with thermal overloads, have been in normal operation for more than five months and have not tripped off since I sealed them with the tape.

Since taping the fixtures was so time consuming—it took roughly 30 to 45 minutes per fixture—not to mention the frustration of trying to reattach the trim with the small springs, I decided to try manufactured retrofit kits. To analyze the air sealing effectiveness of the manufactured trims, I conducted a blower door/pressure pan test on a recessed-can frame that was insulated

with fiberglass batt insulation. The area where the fixture was installed was depressurized to -50 Pa. I did a smoke test at each fixture to identify air leakage locations. A standard, open black baffle trim indicated a pressure pan reading of -48 Pa, which I used as a baseline. The pressure pan test was done first with the standard trim and then again after the retrofit trim was installed (see Table 3). The installation time for all the retrofit kits was approximately ten minutes each. I liked the large springs on the retrofit trims that mounted the trim to the ceiling. Less frustration is also a good thing.

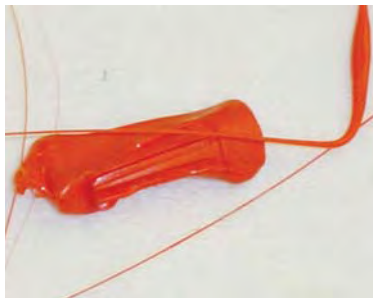
Can Man's Latest Wisdom

Air leakage through recessed-can lights is still a problem and I expect this problem to continue in the future. Although the manufactured trims or retrofit kits I tested indicated some reduction in air flow during the blower door and pressure pan test, they shouldn't be relied on solely to stop this flow of air. Most of the kits were successful in reducing the amount of air leakage inside the fixture. However, most did not address the leakage associated with the trim and the drywall connection, where 50% or more of the leakage occurs. The cost of

the retrofit kits made no difference in how well they reduced air flow. The Lithonia Air-Tite foam tape kit, at a cost of \$2.35, did as well as the Builders Best retrofit kit, at a cost of \$34. All the manufactured kits tested were easy to install.

The duct tape manufactured by Nashua Tyco Adhesives was best for both air sealing recessed-can lights inside the fixture and sealing the trim to the drywall. Nashua manufactures a butyl-backed tape with a 5-mil adhesive that may be another solution. However, I did not test this product. Caulking may be another way to air seal the fixtures, although that may get a little messy and caulking material may not bridge larger gaps between the drywall and the fixture. The drawback with taping was the time needed to complete the air-sealing and to reinstall the trim. The problem that added the most time to the air sealing was the small springs that attach the trim to the ceiling—did I mention how annoying they were? Using the large wing type of springs that are on the trims in the manufac-

tured retrofit kits could reduce the installation time. The cost of a roll of tape, which could be used for dozens of



The wire nut that I used to jumper out the thermal cutout switch while testing a 100W A-base lamp in an air-sealed fixture melted off the connection.

fixtures, was less than \$20. The Halo Air Tite Super trim and the duct tape may be another option.

Although the duct tape method of air sealing IC-rated recessed-can lights is not in the International Energy Conservation Code or the National Electrical Code, it appears to be the most successful method

of reducing air flow through IC-rated recessed-can lighting located in inaccessible attic spaces—not to mention keeping the aliens off your roof. If you decide to use this method of air sealing, don't ever use it on non IC-rated lights, make sure to use lower-wattage reflector spotlights or fluorescent spotlights (see "New Reflector CFLs That Can Take the Heat," *HE* March/April '05, p. 9), and make sure that the fixtures have thermal overload protection (IC rated).



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